New Uzbekistan University

Statement of Ethics: I agree to complete this exam without unauthorized assistance.

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Midterm Exam

COURSE NAME:	Design and Analysis of Algorithms	COURSE NUMBER:	CS 212
EXAMINATION DURATION:	120 MINUTES	EXAM VERSION	A
ADDITIONAL MATERIALS ALLOWED TO USE:	No additional materials or items are allowed to be used.		

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST! VIOLATION OF THE RULES CAN LEAD TO A LOSS OF POINTS.

- Read and strictly follow extra instructions at the beginning of each Section.
- Put your signature and write your ID on even pages.
- Follow the University Examination Rules.

FOR INSTRUCTOR USE ONLY (DO NOT WRITE ANYTHING):

Question:	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Points:	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Score:														

Question:	15	16	17	18	19	20	21	22	23	24	25	26	27	Total
Points:	2	3	3	3	3	3	3	3	3	3	3	20	20	100
Score:														

1 Multiple choice questions

Choose the correct answer.

1. (2 points) Which of the following sorting algorithms has the best average-case time complexity?
   A. Bubble Sort
   B. Merge Sort
   C. Insertion Sort
   D. Selection Sort

1. ____________

2. (2 points) Which sorting algorithm is not stable in its standard implementation?
   A. Merge Sort
   B. Bubble Sort
   C. Quick Sort
   D. Insertion Sort

2. ____________

3. (2 points) Which sorting algorithm is not in-place in its standard implementation?
   A. Bubble Sort
   B. Quick Sort
   C. Heap Sort
   D. Merge Sort

3. ____________

4. (2 points) What is the best-case running time of Insertion Sort?
   A. $O(n)$
   B. $O(n \log n)$
   C. $O(\log n)$
   D. $O(n^2)$

4. ____________

5. (2 points) How many of the following algorithms use the divide-and-conquer strategy? Selection Sort, Insertion Sort, Bubble Sort, Merge Sort, Heap Sort, Quick Sort, Counting Sort, and Radix Sort.
   A. 1
   B. 2
   C. 3
   D. 4

5. ____________

6. (2 points) In the worst case, how many comparisons does Quick Sort perform on an array of n elements?
   A. $O(n)$
   B. $O(n \log n)$
   C. $O(\log n)$
   D. $O(n^2)$

6. ________

7. (2 points) What input leads Quick Sort to its worst-case time complexity in the naive pivot choice (last element)?
   - A. Random order array
   - B. Already sorted array
   - C. Reverse sorted array
   - D. Both B and C

7. ________

8. (2 points) Consider the recurrence $T(n) = T(\frac{n}{2} + \sqrt{n}) + 75$. What is the asymptotic complexity?
   - A. $\Theta(\log n)$
   - B. $\Theta(n)$
   - C. $\Theta(n \log n)$
   - D. $\Theta(1)$

8. ________

9. (2 points) Consider the recurrence $T(n) = 2T(\frac{n}{2}) + n \log n$. What is the asymptotic complexity?
   - A. $\Theta(n \log n)$
   - B. $\Theta(n \log^2 n)$
   - C. $\Theta(n^2)$
   - D. $\Theta(n^2 \log n)$

9. ________

10. (2 points) Consider the recurrence $T(n) = T(\frac{n}{2}) + T(\frac{n}{3}) + n + 3\sqrt{n} + 5\log n + 9$. What is the asymptotic complexity?
    - A. $\Theta(n\sqrt{n}\log n)$
    - B. $\Theta(n\sqrt{\log n})$
    - C. $\Theta(n\sqrt{n})$
    - D. $\Theta(n)$

10. ________

11. (2 points) Counting Sort works best when:
    - A. Input is large, but k is very large compared to n
    - B. Input is large, and $k = O(n)$
    - C. Input is small, but k is very large
    - D. Input is floating-point numbers in arbitrary range

11. ________

12. (2 points) In a max-heap with height 5, the third largest element can only be located among how many nodes?
    - A. 2
    - B. 4
    - C. 6
    - D. 8

12. __________

13. (2 points) What is the time complexity of extract-max in a max-heap with n elements?

   A. $O(1)$
   B. $O(\log n)$
   C. $O(n)$
   D. $O(n \log n)$

13. __________

14. (2 points) Suppose we apply the bucket sort algorithm with n buckets to sort n numbers that are drawn uniformly at random from the interval $[0, 1)$. On expectation, how many buckets will remain empty?

   A. n/e
   B. $n/2$
   C. $\sqrt{n}$
   D. 0

14. __________

15. (2 points) You want to maintain the running median of a sequence as new numbers are inserted dynamically. A common approach uses two heaps: a max-heap to store the smaller half of the elements and a min-heap to store the larger half. What is the worst-case time complexity for inserting a new element and performing any necessary rebalancing?

   A. $O(1)$
   B. $O(\log n)$
   C. $O(\log^2 n)$
   D. $O(n)$

15. __________

2 Short Answer Questions

Please provide your responses in the spaces provided. Only fully complete answers will be eligible to receive full marks.

16. (3 points) When building a heap from an existing array of n elements, should we use the top-down method or the bottom-up method? Justify your answer in terms of time complexity.

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17. (3 points) The k-th order statistic of an array of n elements can be found in different ways. Analyze the running time for each of the following three approaches:

    - Sorting the array and then picking the k-th element.
    - Building a min-heap and extracting elements until the k-th is reached.
    - Using the deterministic Select (Median-of-Medians) algorithm.

    State the time complexity for each method in terms of n and k, and briefly justify your answers.

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18. (3 points) An inversion in an array $A[1 \ldots n]$ is a pair of indices (i, j) such that $i < j$ and $A[i] > A[j]$. For an array of size 10, what is the maximum number of possible inversions?

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19. (3 points) Explain how the number of inversions affects the running time of Insertion Sort.

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20. (3 points) Let $f(n)$ be an asymptotically non-negative function satisfying the following conditions:

- $f(n) = o(a^n)$ for every constant $a > 1$, and
- $f(n) = \omega(n^k)$ for any positive integer k.

Provide an example of such a function $f(n)$.

21. (3 points) Decide whether the statement $f(2n) = O(f(n))$ holds for every asymptotically nonnegative function $f(n)$. Either prove the statement (give a correct proof for all such f), or disprove it by giving a counterexample.

22. (3 points) Solve the recurrence $T(n) = 3T(\frac{n}{2}) + n^2$.

23. (3 points) You are given two linked lists A and B, each containing a subset of integers from 1 to n. You are asked to design an $O(|A| + |B|)$ algorithm to return a new linked list that represents the intersection $A \cap B$. How would you approach this problem?

24. (3 points) Aziz and Dilshod are playing a game with a given array A of integers. Aziz secretly picks two indices i and $j (i \leq j)$ and asks Dilshod to quickly tell him the sum of all elements of the input array A between positions i and j, that is, $\sum_i^j A[i]$. Dilshod is allowed to preprocess the array in-place before the game starts. What preprocessing technique should Dilshod use so that each query is answered in $O(1)$ time?

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25. (3 points) You are given an input array of size $n-1$, where each element is a distinct integer between 1 and n. Hence, exactly one number in that range is missing from the array. Explain an in-place algorithm that runs in $O(n)$ time to find the missing number.

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3 Algorithm Design Questions

Please provide your responses under the problem statements. To receive full marks, you need to provide proof of correstness and running-time analysis of your algorithms.

26. (20 points) Sevinch is analyzing the daily stock prices of a company, stored in an array $A[0...n]$, where each element represents the stock price on a given day. He wants to find the best days i and $j(j > i)$ to buy and then sell the stock, so that the profit $A[j] - A[i]$ is maximized. Design an algorithm that finds such indices in $O(n \log n)$ time.

27. (20 points) The Intergalactic Science Council is selecting a team of 3 astronauts for a prestigious competition. There are n candidates, each with an integer skill score stored in the array A. To form a perfectly balanced team, the sum of the three selected candidates' skill scores must equal a given target s. Given A and s, your task is to determine whether there exists a subset of exactly three candidates whose sum equals s. Your algorithm should run in $O(n^2 \log n)$ time.